

Name: _____



Statistical problems - statistical summaries

Task A: Problems with statistical measures based on the data

1) The four sets of data all have a mean of seven.

In which sets of data do you think that this average provides a good summary of the information? Give reasons for your answers.

a) 1, 1, 1, 1, 1, 1, 43

b) 7, 7, 7, 7, 7, 7, 7

c) 1.5, 2.8, 6.1, 6.9, 7.9, 9.4, 14.4

d) 4, 5, 6, 7, 8, 9, 10

2) Find the mean, median and mode for the set of data.

1, 2, 3, 3, 4, 4, 5, 6, 9, 9, 9

Which average least represents the central tendency for this data? Explain why.

Task B: Choosing statistical measures based on the context

1) The table contains data from The Met Office about the weather in Cambridge in 2004. It shows the number of days in each month where there was frost.

month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
no. days	7	11	3	0	0	0	0	0	0	0	3	9

- a) Find the mean number of days of frost in a month, giving your answer to 2 decimal places.
- b) Find the median number of days of frost in a month.
- c) Find the mode.
- d) Which measure of average would be the least useful for describing the amount of frost that could be expected during a typical month in Cambridge? Explain why.

2) A homeowner is selling their house. The data below shows the prices for the last five houses that were sold on the same street.

£150 000 £350 000 £360 000 £350 000 £210 000

- a) Find the mean and median house price.
- b) Which average suggests that houses here are worth more money?
- c) Which average suggests that houses here are worth less money?
- d) Which average might the seller prefer to use?
- e) Which average might the buyer prefer to use?

